MPP	MULTIDISCIPLINARY POLICY AND PROCEDURE

TITLE:	MEDICATION ADMINISTRATION
DEPARTMENT:	PHARMACY
APPLIES TO:	PHARMACIST, DOCTORS, AND NURSES

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1. PURPOSE

1.1. To provide written guideline on the provision of safe medication administration and documentation in timely manner.

1.2. To ensure patient's safety by following the eight (8) rights of medication administration and by monitoring adverse effect of the medication.

1.3. To prevent errors in medications.

2. DEFINITION AND ABBREVIATION

2.1. **Medication** – is a substance administered for the diagnosis, cure, treatment, or relief of symptoms or for prevention of disease.

2.2. **Medication Administration** – is giving a specified medication to the patient according to the physician's order and following established policies and procedures.

2.3. **Look Alike Sound Alike Medications** – are those medications which are look-alike or sound-alike and may lead to confusion which may result in error dispensing and administering medications.

2.4. **High Alert Medications** – are those drugs that are treated with extra precaution in order to avoid the inadvertent use in their undiluted forms and to avoid inappropriate dosing and administration.

2.5. **Medication Administration Records (MAR)** – used to record information about the drugs a patients receives under a prescriber's orders.

2.6. **Intravenous (IV) push** – is a specific amount of medication (diluted or undiluted) given directly into the vein over 30-60 seconds, it should be given by trained registered nurse.

2.7. **NPO** – (nil per os) a Latin phrase meaning nothing by mouth. It is a medical instruction to withhold oral food and fluids from a patient for various reasons.

2.8. **PRN** – (pro re nata) a Latin phrase meaning in the circumstances or as the circumstance arises; commonly used in medicine to mean as needed or as the situation arises.

2.9. **Inotropes** -medicines that change the force of heart's contractions.

3. RESPONSIBILITY

3.1. **Director of Pharmacy** ensures that pharmacist and nursing staff were oriented and adheres to this policy.

3.2. **Pharmacist** ensures that medications dispensed to nursing units are not expired, properly labelled and integrity (general appearance) maintained.

3.3. **Prescribing Physician**

3.3.1. Responsible to write prescription with the following:

3.3.1.1. complete addressograph of the patient

3.3.1.2. date of the prescription

3.3.1.3. the medication

3.3.1.4. the route

3.3.1.5. the frequency

3.3.1.6. the dose

3.3.1.7. the strength

3.3.1.8. diagnosis

3.3.1.9. dosage form

3.3.2. Preferably should prescribe in generic name unless there is bioavailability or brand name is recommended (such as combination products) or indicated for safety reasons (such as look-alike and sound-alike drug names).

3.3.3. Should avoid unnecessary use of decimal points.

3.3.4. Should indicate the number of treatment days.

3.3.5. Should specify the strength and quantity.

3.3.6. Should only use hospital's approved abbreviations.

3.3.7. Should utilize the standard medication administration time as per hospital's policy.

3.4. **Qualified Registered Nurses and Midwives**

3.4.1. Responsible for administering medications within their scope of practice.

3.4.2. Are knowledgeable about the effects, side effects and interactions of medications and take actions as necessary.

3.4.3. Determine all orders for an individual are clear, complete, current, legible, and appropriate for the client before administering any medication.

3.4.4. Act upon verbal and telephone orders only in emergent and urgent situation except for narcotic and controlled drugs wherein telephone order is NOT ALLOWED at all.

3.4.5. Administer only medications they themselves or a Pharmacist have prepared, except in an emergency.

3.4.6. Educate patient and family about their medications.

3.4.7. Verify that medications orders, pharmacy labels and/ or medication administration records are complete and include the name of the client, name of medication, medication strength and dosage, route, frequency with which the medication is to be administered.

3.4.8. When a medication error or near miss occurs at any point in the process of prescribing,

compounding, dispensing or administering a medication, nurses take appropriate steps to resolve and report it in a timely manner.

3.5. The **Head Nurse** ensures that this policy is strictly implemented in their respective area.

3.6. The **Nurse Educator** ensures that nurses and trained midwives are competent in medication administration.

4. POLICY

4.1. Only registered nurses and trained midwives with appropriate qualification and experience who successfully completed and passed the medication administration competency are allowed to administer medication with or without supervision.

4.2. Registered nurse/ midwife must know the first basic eight (8)rights of medication at all times:

4.2.1. Right Patient

4.2.2. Right Drug

4.2.3. Right Dose

4.2.4. Right Frequency

4.2.5. Right Route

4.2.6. Right Time

4.2.7. Right Documentation

4.2.8. Right Response

4.3. All medications administered to patients shall have a Physician's order and nurses should expect that the order shall be dated and timed and clearly state the medication's name, dosage, frequency, route, duration and any special instruction.

4.4. Medication room shall only be accessible to authorized personnel hence; it should be locked at all times with senior nurse in-charge of the key.

4.5. The nurse must assess for any allergies to medications or ensure drug allergy is documented in the patient's file.

4.6. Administration of medications to female patients that involves exposure or touching of the body must be carried out by a female nurse except in cases of extreme emergency.

4.7. Medications brought to the hospital by the patient shall not be administered or left with the patient without the written order of the Physician for continuation of any such drug. This medication shall be sent to Pharmacy as per policy refer to **Patient Own Medication Policy and procedure.**

4.8. Near expire drugs shall be separated in a designated area in their respective unit and given priority for administration.

4.9. NPO Status: If the Physician writes "NPO" on the Physician's order the nurse shall clarify with the physician about which, if any oral medications should be given.

4.10. Narcotics are to be checked on every shift by the responsible nurse. All narcotics and controlled drugs will be kept under double lock system.

4.10.1. The charge nurse is responsible for the narcotic key throughout the shift.

4.11. High Alert Medications, Narcotics and Controlled are double checked in preparation and administration.

4.12. Since all pre-operative medications orders are cancelled when a patient goes to surgery, the nurse should expect that each post-operative medication will be separately and newly ordered.

4.13. **General Consideration**

4.13.1. Nurses must be able recognize the limits of their own knowledge and skills as under the law, nurses are responsible for their own actions regardless of whether there is a written order.

4.13.2. Nurses should question any order that appears to be unreasonable and refuse to give the medication until the order is clarified.

4.13.3. The nurse needs to be aware of developmental factors:

4.13.3.1. Pregnant women must be careful about taking any medication especially in their first trimester because of the possible adverse effects on the fetus.

4.13.3.2. Infants require smaller doses because of their body size and the immaturity of their organs.

4.13.4. Diet may contain nutrients that can interact with medications and increase or decrease action.

4.13.5. Time of administration is important because medications are absorbed more quickly if the stomach is empty; however, some medications irritate the gastrointestinal tract and are given after meal.

4.14. **Verification of Physician's Order**

4.14.1. It is within acceptable professional nursing practice for nurses to seek verification of a physician's order before administering medication, if their concerns are based on clinical scientific knowledge and/ or standards of practice.

4.14.2. Whenever an order needs verification, the nurse shall consult with the head nurse or charge nurse and discuss the concerns. This will provide a second nursing opinion and possibly settle the issue at this point.

4.14.3. If the order is still not acceptable, the nurse and/ or the head nurse should contact the ordering Physician and discuss their concerns with him/ her.

4.14.3.1. The nursing supervisor, the Nursing Director must also be notified if the issue remains unresolved and requires further action.

4.14.3.2. Additional specialist pertinent to issue (Pharmacist) may be consulted as needed.

4.14.3.3. Pharmacy should be informed of the concerns surrounding the order, and will hold the questionable orders until the situation is resolved.

4.15. **Medication Preparation**

4.15.1. Prior to administration, check the medication sheet against Physician's orders. Ensure the correct medication has been selected based on the medication order and label, check for particulates or discoloration. Ensure medication not expired.

4.15.2. Medications must be prepared for one patient at a time. Prior to medication preparation, the nurse will perform hand hygiene. Medication preparation area should be clean, have good lighting, and located in a closed area to avoid distraction.

4.15.3. Medication storage areas are locked at all times except when nurses are preparing medications.

4.15.4. Unit dose packages must be opened at the bedside. Medications that require preparation, i.e. measuring, crushing or bulk medication must be prepared in the medication preparation room.

4.15.5. Medications are not to be left on trays, bedside tables, etc.

4.15.6. The nurse should not administer medication prepared by the other nurse.

4.15.7. A nurse is not to pour a medication from one bottle to another. A nurse is not to put a medication back into a bottle.

4.15.8. Nurses/ midwives are not authorized to re-label medications bottles or containers. Medications which loses its label or which lacks a legible label must never be administered to the patients. The Head Nurse/ designee must inform the Pharmacy and secure the medication until a replacement medication dose received from the pharmacy.

4.15.9. All medications requiring refrigeration should be identified by the Pharmacy Department and place in medication refrigerator.

4.15.10. Each nurse should know the action, indication, normal dosage range and route, adverse effect, contraindication, drug interactions, nursing considerations and how to compute the dosage. Drug Information Manuals and Hospital Formulary are available in each unit. If uncertain, the nurse can consult the Pharmacy Department.

4.15.11. The nurse should assess as well the following but is not limited to in accordance to the drugs to be administered:

4.15.11.1. Vital Signs:

4.15.11.1.1. Blood pressure for anti-hypertension

4.15.11.1.2. Pulse for cardiac drugs like digoxin

4.15.11.1.3. Pain score for analgesia

4.15.11.1.4. Temperature for anti-pyrexia

4.15.11.2. Laboratory Results:

4.15.11.2.1. CBC, Chemistry, LFT, Coagulation Profile, etc.

4.15.11.3. Drug Levels of the following

4.15.11.3.1. Digoxin

4.15.11.3.2. Phenytoin

4.15.11.3.3. Theophylline

4.15.11.4. Electrolytes

4.15.11.4.1. Know level before administration

4.15.11.5. Blood Glucose before insulin or Oral Hypoglycemic Agents

4.16. Administering Medications

4.16.1. At the patient's bedside, verify the patient's name and medical record number on the medication sheet and identification band.

4.16.2. The nurse administering the medication should follow the first eighth (8) basic rights for medication administration.

4.16.3. Check for contraindications to patient receiving the medication i.e. allergies.

4.16.4. The nurse shall assess the patient's ability to swallow medication and remain with the patient until the medicine is being swallowed.

4.16.5. High alert medications shall be checked utilizing independent double-checking, and signed by two (2) Registered Nurses before administering to patient i.e. Heparin, Digitalis, Coumadin, Warfarin, Insulin.

4.16.6. Two (2) nurses must double-check with each other the calculation of the dosage of high alert medication and both nurses must sign in the Medication Administration Record.

4.16.7. All controlled drugs and narcotics must be checked and signed by two (2) staffs, one of whom should be a charge nurse, prior to administration.

4.17. Standard Medication Administration Time

4.17.1. **"Stat"** orders will be given immediately and only once upon receipt of the medication

on the unit. The allowance of thirty (30) min is given from the time the order has been written to the preparation and delivery of the medication to the ward.

4.17.2. **"Single or one-time"** order is for medication to be given once at a specified time.

4.17.3. **"Standing or Routine"** Order may or may not have any termination date, may be carried out indefinitely until an order is written to cancel it, or may be carried out for a specified number of days.

4.17.4. **"Loading dose"** are initial large dose of a drug given to bring blood, tissue or fluid level to an effective concentration quickly.

4.17.5. **"PRN or as needed"** order permits the nurse to give the medication to patients that requires the medication upon the discretion of the nurse.

4.17.6. Medication may be given 30 minutes before or after the scheduled time according to established Medication Administration Time.

4.17.7. Standard Administration Time should be used for dosing schedule when possible in order that the nursing units and pharmacy schedule are coordinated and ensure that dose will be available when needed and wasted dose will be avoided unless specified by the physician or indicated. (see attachment)

4.17.8. Be aware that insulin dose must be given based on patient's meal time, prescribed dietary plan and prescribed by physician.

4.17.9. All medications are automatically placed on the standard schedule attached except for Intravenous medications wherein:

4.17.9.1. Intravenous medications are placed on the schedule closes to the time the order is written or

4.17.9.2. To the closest schedule which does not already have two (2) Intravenous medications assigned to it.

4.17.10. Four (4) times daily (QID) and every six (6) hours (Q 6 hours) may be implemented as either primary or alternate time based on patient sleep patterns.

4.17.11. Alternative times and exceptions are based on the patient's clinical situation and are negotiated between the nurse and the pharmacist based on the following conditions:

4.17.11.1. When specific regimens, guidelines or clinical indications requires deviations from the standard time.

4.17.11.2. For strong patient preference; when a patient is on a different time schedule at home and a hospital change would create a hardship for the patient returning home.

4.17.12. The Standard Administration Time should be used except for the following:

4.17.12.1. Intensive Care Patients receiving more than two (2) medications can be re-scheduled.

4.17.12.2. Emergency admissions can have required medications as stat dose and regularize in next dose.

4.17.13. Warfarin dose to be given at 1700H unless otherwise prescribed.

4.17.14. Antibiotics are to be started upon receiving the order and the Medication from the pharmacy if not available as floor stock.

4.17.15. The in-charge nurse will decide the administration times until standardized medication sequence order has been reached.

4.17.16. Orders for drug other than antibiotics, receive more than one (1) hour after their scheduled time for administration will be started at the next scheduled times unless ordered otherwise.

4.17.17. An initial dose may be ordered to be given off schedule by requesting a "now" or "today" and specifying a standard schedule.

4.17.17.1. Example: Digoxin 0.25mg today then BID or Nifedipine SR 20mg now then Q12 hours.

4.17.18. If possible, the last dose of the day for diuretics should be given not later than 1800H.

4.17.19. Insulin timing will be specified by the physician.

4.17.20. If Oral Hypoglycaemic Agents have different divided doses, the schedule must be specified by the physician.

4.18. **Intravenous (IV) Push Medication**

4.18.1. Rapid IV push is a specific amount of medication given directly into the vein over less than 30 seconds.

4.18.1.1. Exception: adenosine is given rapid push at least within 3-5 seconds proximal to the heart because of its extreme short half-life.

4.18.2. Slow IV push is a specific amount of medication given into the vein over 3-5 minutes.

4.18.3. Any medication administered for greater than 5 minutes must be given by soluset.

4.18.4. Intravenous Medications given in central venous line must be flushed with at least 20 milliliters of normal saline and at least 10 milliliters with intravenous cannula.

4.18.5. Inotropes must be administered via a central venous catheter. Blood pressure and abnormal heart rhythms (arrhythmias) are being monitored closely.

4.18.6. A Standard Calculation of Common Drugs and Inotropes used in nursing unit as continuous infusion are provided. (see attachment)

4.20. For any Medication Administration Error, refer to **Medication Error Policy Procedure.**

4.21. Nurse competency assessment for drug administration shall be conducted and included in the personnel files.

5. PROCEDURE

5.1. Review medication sheet at the beginning of each shift and frequently for changes during the shift.Check for:

5.1.1. Day/ Month/ Year and time order was written.

5.1.2. Patient's allergies

5.1.3. Automatic stop order

5.1.4. Medical record number

5.1.5. Medication dose, route, frequency, time and duration

5.2. Check medication label against medication record for:

5.2.1. Correct medication

5.2.2. Correct dose

5.2.3. Expiration date

5.3. Wash hands before medication administration to prevent contamination.

5.4. While preparing medications, check again the label for the medication name and dose.

5.5. Take medication and medication sheet to patient room.

5.6. Medication are taken to the patient's room and opened at the bedside.

5.7. Medications that needs preparation i.e. measuring, crushing, or bulk medications maybe prepared at the medication cart.

5.8. Do not take patient medication sheet into the room if patient is in isolation room. If applicable, stamp the patient's addressograph/ patient's identification on a piece of paper and take into the patient's room for identification.

5.9. Check medical record number and name on wristband against the medication sheet. Ask patient his/ her name and check medication name, dose, route, and time.

5.10. Explain procedure to the patient to relieve anxiety and gain cooperation.

5.11. Remain with the patient during the oral medication to ensure that medication has been taken.

5.12. If patient refused to take medication, return unopened medication on the cassette.

5.13. If medication was opened and refused, waste in sink and have another senior nurse witness and sign wastage on medication sheet to serve as a witness that the medication was discarded appropriately.

4.18.7. Guideline for Medications that can be Safely Administered Intravenous Push is provided as well. (see attachment)

4.18.8. Monitoring of Vital Signs:

4.18.8.1. Baseline vital signs (blood pressure, oxygen saturation, pulse and respirations) will be obtained before an IV push of narcotic, controlled and sedation agents is administered and every 15 minutes after administration for at least one hour.

4.18.9. Borrowing of medications from another patient's supply or from the Crash Cart is not allowed except in a life-threatening situation, in which case it shall be replaced immediately.

4.19. **Charting of Medications**

4.19.1. Copying or transcribing of Physician's order is NOT ALLOWED. The responsible staff nurse/ midwife will inform the physician to re-write the order on the medication sheet as needed.

4.19.2. Document on the medication sheet the medication administered at the bedside.

4.19.3. Missed dose due to diagnostic procedures:

4.19.3.1. When a medication time must be changed because of diagnostic procedures, etc., the nurse shall consult the pharmacy as needed for missed medications doses and rearrange the medication schedule.

4.19.4. Withholding medications

4.19.4.1. Any medication that is withheld shall be initialed on the medication sheet and write not given.

4.19.4.2. The nurse/ midwife shall document in the nurse's notes the reason the drug was not given and, if applicable, physician was notified.

4.19.5. Discontinued medication – when a drug is discontinued, the nurse/ midwife shall:

4.19.5.1. Ensure the physician write discontinued next to the last dose given.

4.19.5.2. To discontinue a regular or a PRN drug, close appropriate section by oblique line and sign your name.

4.19.6. PRN medications:

4.19.6.1. PRN medications shall be documented on the medication sheet with the date, time, nurse's name, and signature.

4.19.6.2. The patient's response to PRN medications is written in the nurse's notes.

4.19.7. On discharge, the medication sheet is included in the patient's file before sending to the Medical Record Department.

5.14. For IV preparation, if the medication has been diluted and there is wastage, ALWAYS discard unused diluted portion of the prepared IV medication immediately before going to the bedside. Dispose the empty ampules and sharp needles in the sharp container.

5.15. Record medications given in the medication sheet.

5.15.1. Time medication is given.

5.15.2. Signature of the nurse.

5.16. Encircle initials of staff nurse in time slot for medications held or not taken. Record in the comment section the drug, date, time, reason and initial of staff nurse.

5.17. Dispose sharps in the sharp disposal container to avoid needle stick/ sharps injury.

5.18. Record patient's response to PRN medications.

5.19. Document and sign.

6. MEASUREMENT

6.1. 100% compliance of all nurses and midwives with the policy.

6.2. 100% of patients will be safe and secure by implementing this policy.

7. REFERENCES / CROSS-REFERENCES

7.1. CBAHI Standard 3rd Edition.

7.2. Policy and procedure in Medication Administration, APP-QM-0001.

7.3. Medication Administration Policy, MPP-MM-0037.

7.4. Medication Storage, Preparation, and Administration Policy, DPP-NR-NP-046-047-048.

7.5. Smith, S.F., Duell, D.J., Martin, B.C. (2004) Clinical Nursing Skills, Basic to Advanced, 6th Ed. New Jersey: Prentice Hill.

7.6. Timby, B.K. (2003) Fundamental Skills and Concepts in Patient Care, 7th Ed. Philadelphia: Lippincott.

8. ATTACHMENTS

8.1. Doctor's Order Sheet.

8.2. Medication Sheet.

8.3. Standard Medication Administration Time.

8.4. Standard Calculation of Common Drugs and Inotropes.

8.5. Guidelines for Medication Safely Administered Intravenous Push.

9. REVISION CLAUSE

9.1. Revision in logo, code, font, signatures, and date

9.2. Changes in policy contents.